Lab Exercise 1

Perform the following operations:

a) Log in to the system
b) Use vi editor to create a file called myfile.txt which contain some text.
c) Correct typing errors during creation
d) Save the file
e) Logout of the file

Solution:

$ login: <user name>
$ password: ******
$ vi

~ Unix is Case Sensitive

~ Never leave the Computer without logging out when you are working in a time sharing or network environment.

Type <Esc>

: wq myfile

$

Lab Exercise 2

Write a shell script to perform swapping of 2 no. using 3rd variable

Coding:

```
#!/bin/bash
echo a
read a
echo b
read b
c=$a
a=$b
b=$c
echo a=$a
echo b=$b
```

Lab Exercise 3

Write a shell script to perform swapping of 2 no. without using 3rd variable

Coding:

```
#!/bin/bash
echo a
read a
echo b
read b
a=`expr  $a + $b`
b=`expr  $a - $b`
a=`expr  $a - $b`
echo a=$a
echo b=$b
```

Lab Exercise 4

Write a shell script to perform all arithmetic operator of decimal value

Coding:-

```
#!/bin/bash
a=10.2
b=2.3
echo a=$a
echo b=$b

add=`expr $a + $b | bc`
echo add=$add

sub=`expr $a - $b | bc`
echo sub=$sub

mult=`expr $a \* $b | bc`
echo mult=$mult

div=`expr $a / $b | bc`
mod=`expr $a % $b | bc`
echo mod=$mod
```

Lab Exercise 5

Write a shell script to find the smallest no. in 2 no. using conditional operator

Coding:-

```
#!/bin/bash
echo a
read a
echo b
read b
if test $a –lt $b
then
echo a<=$a
else
echo b<=$b
fi
```

Lab Exercise 6

Write a shell script to find largest among 3 numbers

Coding:-

```
#!/bin/bash
echo a
read a
echo b
read b
if test $a -gt $b –a $a –gt $c
then
echo a>=$a
else
if test $a -gt $b –a $a –gt $c
then
echo b>=$b
else
then
echo c>=$c
fi
fi
```

Lab Exercise 7

Write a shell script to print a table of given no. using while loop statement

Coding:-

```
#!/bin/bash
echo a
read a
b=1

while test $b –le 10
do

c=`expr $a \* $b`
echo $a * $b=$c
b=`expr $b + 1`

done
```

Lab Exercise 8

Write a shell script to print 1 to 10 no. in series using while loop statement

Coding:-

```
#!/bin/bash
a=1
read a

while test $a –le 10
do

echo $a
a=`expr $a + 1`

done
```

Lab Exercise 9

Write a shell script to print a given no. in reverse order

Coding:-

```
#!/bin/bash
echo a
read a
rev=0
while test $a –ne 0
do
temp=`expr $a % 10`
rev=`expr $rev \* 10`
rev=`expr $rev + $temp`
a=`expr $a / 10`

done
echo $rev
```

Lab Exercise 10

Write a shell script to show the no. is in palindrome or not

Coding:-

```
#!/bin/bash
echo a
read a

while test $a –ne 0
do

temp=`expr $a % 10`
rev=`expr $rev \* 10`
rev=`expr $rev + $temp`
a=`expr $a / 10`

done
echo $rev
if test $rev –eq $a
then
echo the no. is palindrome
else
echo the no. is not palindrome
fi
```

Lab Exercise 11

WAP to perform all the arithmetic operators using case structure

Coding:-

```
#!/bin/bash
echo a
read a
echo b
read b
echo enter ur choice
echo 1=add,2=sub,3=mult,4=div,5=per
read num
case $num in
1) echo sum is=`expr $a + $b`
2) echo sub is=`expr $a - $b`
3) echo mult is=`expr $a \* $b`
4) echo div is=`expr $a / $b`
5) echo per is=`expr $a % $b`
*) echo wrong choice;;
esac
```

Lab Exercise 12

WAP to calculate the average marks & its percentage and find grade

Coding:-

```
#!/bin/bash
echo enter marks
echo a
echo b
read a
read b
echo c
read c
d=`expr $a + $b + $c`
echo d=$d
e=`expr $d /* 100`
f=`expr $e / 300`
echo f=$f
if test 4f –fgt 60
then
echo first division=$f
else
if test $f -gt40 –a $f –et 60
then echo second div.=$f
else
if test $f –et 40
then
echo 3rd=$f
fi
fi
fi
```

Lab Exercise 13

WAP to print the pattern shown

```
*
**
***
****
```

Coding:-

```
#!/bin/bash
echo n, red n
for((i=1;i<=4;i++))
do
for((j=1;j<i;j++))
do
echo –n "*"
done
echo
done
```

Lab Exercise 14

Aim:- WAP to print the pattern shown

```
****
***
**
*
```

Coding:-

```
#!/bin/bash
echo n, red n
for((i=4;i>=1;i--))
do
for((j=1;j<=i;j++))
do
echo –n "*"
done
echo
done
```

Lab Exercise 15

WAP to print the pattern shown

```
   *
  **
 ***
****
```

Coding:-

```
#!/bin/bash
echo m
read n
for((i=1;i<=4;i++))
do
for((j=m;j>=1;j--))
do
if test $j –gt $i
then
echo –n "*"
else
echo –n " "
fi
done
echo " "
done
```

Lab Exercise 16

WAP to print the pattern shown

```
      *
    * *
  * * *
* * * *
```

Coding:-

```
#!/bin/bash
for((i=1;i<4;i++))
do
for((j=0;j<4-i;j++))
do
echo –n " "
done
for((k=4;k>=4-i;k++))
do
echo -n" * "
done
echo " "
done
```